HomeCare Connect

Database design: ER diagram and every table and column

PostgreSQL database, read from the running app on 30 Oct 2026 (migrations V1 to V14). Sample values are made up; no real people's data is in this document.

What is inside

- **Page 2**: the ER diagram, showing how the 11 tables are linked.
- **One page for each table**: every column with its type, whether it may be empty, its key, default, what it holds and an example value, followed by the links, rules and indexes of that table.
- **Last page**: the allowed words for the columns that hold a fixed set of values (roles, statuses, job types and so on).

The 11 tables at a glance

Table	What it stores	Linked to
`users`	Every account: caregivers, care seekers and admins.	-
`caregiver_profiles`	The extra details of a caregiver: job, address, experience, on/off switch and whether families can see them.	users
`caregiver_rates`	The charges a caregiver offers (per hour, session, day, full day and night).	caregiver_profiles
`caregiver_documents`	Uploaded NIC and certificates, with the admin's decision on each.	caregiver_profiles, users (the admin)
`document_requirements`	Which document is mandatory for which job, set by the admin.	users (the admin)
`reviews`	A care seeker's rating and comment for a caregiver.	caregiver_profiles, users
`bookings`	A care seeker's request for a caregiver, and the amounts asked for once the visit is completed.	users, caregiver_profiles
`payments`	Each try to pay a request's amount through PayHere, or a payment the admin recorded, and how it ended.	bookings
`payment_receipts`	Bank transfer receipts (screenshot or PDF) sent by care seekers, also emailed to the admin.	bookings
`payment_settings`	The admin's two switches: card payments through PayHere and bank transfer. One row.	users (the admin)
`notifications`	Alerts to the admin about new requests and the payment emails to seekers (email or WhatsApp), with their retry state.	bookings

A twelfth table, `flyway_schema_history`, is created by the migration tool (Flyway) to remember which database versions were applied. It holds no app data, so it is left out.

How the tables connect, in plain words

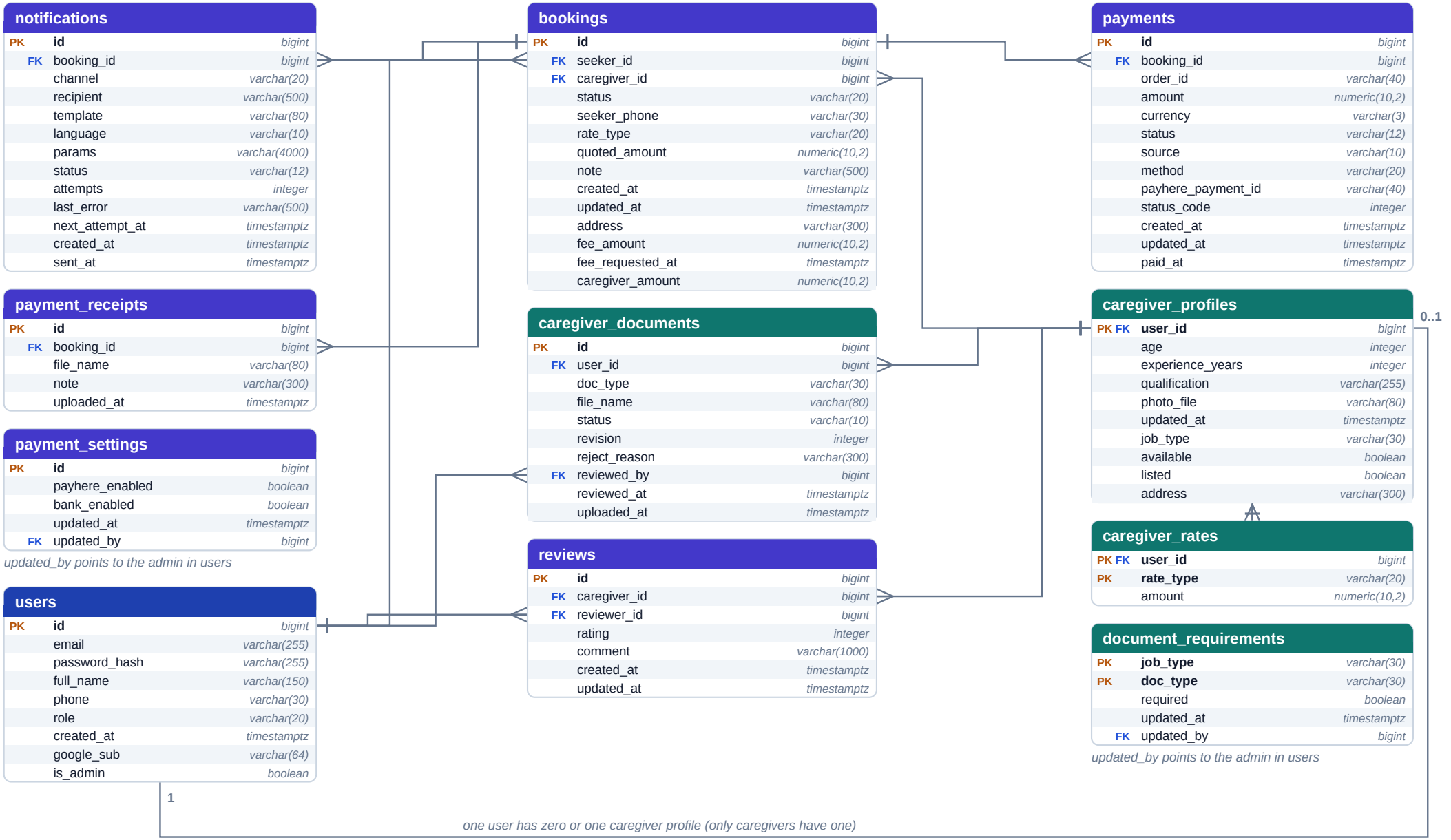
- A person signs up once: one row in **users**. If they are a caregiver, a second row with the same id goes into **caregiver_profiles** (one to one).
- A caregiver has several charges (**caregiver_rates**) and several documents (**caregiver_documents**, at most one per document type).
- Documents are optional until the admin makes one mandatory for a job (**document_requirements**). Then caregivers of that job without it approved are hidden from families.
- A care seeker asks for a caregiver: a row in **bookings** says who asked, whom and where the care is needed. Each request gets an alert row in **notifications**.
- A completed visit gets its amounts on the **bookings** row; payment tries are rows in **payments**, bank receipts in **payment_receipts**.
- A care seeker can review each caregiver once: **reviews**. Writing again replaces the first review.
- Deleting a user deletes everything connected to them (this is called cascade). The one exception: the admin named on a document becomes empty instead.

Words used

PK primary key: identifies a row. **FK** foreign key: points to a row in another table. **UQ** unique: no two rows may share the value. **Empty allowed**: the column may have no value (NULL). A fixed-word column such as status is plain text in the database; only users.role is also checked by the database itself, the others are checked by the Spring Boot application.

ER diagram: how the tables connect

Each box is a table, each line links a foreign key column to the column it points to.



People | Caregiver details | Requests, payments, alerts and reviews | PK primary key | FK foreign key (points to another table) | one to many: one row on the bar side, many rows on the three-pronged side

1. users

Every account in the app: caregivers, care seekers and admins. Logging in, the role and the admin flag all live here.

Column	Type	Empty allowed?	Key	Default	What it holds	Example value
id	bigint	No	PK	auto number	Unique number of the account. The database numbers new rows by itself.	12
email	varchar(255)	No	UQ	-	Login email. Must be unique.	sam@example.com
password_hash	varchar(255)	Yes	-	-	BCrypt scramble of the password; the real password is never stored. Empty for people who only use Google sign-in.	$2a$10$N9qo8uLO...
full_name	varchar(150)	No	-	-	Name shown to other people.	Nina Fernando
phone	varchar(30)	Yes	-	-	Optional phone number typed on the sign-up form.	+94 77 123 4567
role	varchar(20)	No	-	-	What kind of account this is. The database only accepts the two values listed on the last page.	CAREGIVER
created_at	timestamptz	No	-	now()	When the account was created. Filled in by the database.	2026-10-28 09:15:00+05:30
google_sub	varchar(64)	Yes	UQ	-	Google's own id for the person, saved when they sign in with Google. One Google identity can belong to one account only.	109876543210123456789
is_admin	boolean	No	-	false	True for admins. Set only with the make-admin command, never from the website or the API.	false

Links, rules and indexes

- **Primary key**: id.
- No two accounts can use the same email.
- One Google identity can be linked to only one account.
- role must be CAREGIVER or CARE_SEEKER (the only fixed-word column the database checks by itself).

2. caregiver_profiles

The extra details of a user who is a caregiver. There is one row per caregiver and it uses the same id as the user. The 'listed' flag decides whether care seekers can see the caregiver at all.

Column	Type	Empty allowed?	Key	Default	What it holds	Example value
user_id	bigint	No	PK, FK	-	Same number as users.id. Both the primary key and the link to the user.	12
age	integer	Yes	-	-	Optional. Between 18 and 100.	34
experience_years	integer	Yes	-	-	Optional. Years of experience, 0 to 60.	8
qualification	varchar(255)	Yes	-	-	Optional free text: certificates and training, separated by commas.	Nursing diploma, First aid certified
photo_file	varchar(80)	Yes	-	-	Optional. Name of the profile photo file on the server's disk (folder uploads/photos), not the picture itself.	3f2a9c1e-...-b7.jpg
updated_at	timestamptz	No	-	-	When the profile was last changed.	2026-10-28 10:02:11+05:30
job_type	varchar(30)	Yes	-	-	The work the caregiver does. Empty until they choose. Allowed words are on the last page.	NURSING
available	boolean	No	-	true	The caregiver's on/off switch. Off hides them from care seekers.	true
listed	boolean	No	-	false	True while the caregiver has every document the admin made mandatory for their job approved. When nothing is mandatory it is always true. Only listed caregivers are shown to care seekers.	true
address	varchar(300)	Yes	-	-	Where the caregiver lives. Required in the profile form; only the admin sees it. Empty for profiles saved before it was asked.	12 Temple Road, Jaffna

Links, rules and indexes

- **Primary key**: user_id.
- **user_id** points to **users.id**. Deleting that row also deletes this one.
- age, when given, must be from 18 to 100.
- experience_years, when given, must be from 0 to 60.

3. caregiver_rates

The charges a caregiver offers. One row for every way of charging (per hour, per session, per day, full day and night).

Column	Type	Empty allowed?	Key	Default	What it holds	Example value
user_id	bigint	No	PK*, FK	-	The caregiver (users.id). Together with rate_type it forms the primary key.	12
rate_type	varchar(20)	No	PK*	-	The way of charging. Allowed words are on the last page.	DAILY
amount	numeric(10,2)	No	-	-	Charge in Sri Lankan rupees (LKR). More than 0, at most 2 decimals.	9000.00

Links, rules and indexes

- **Primary key**: user_id, rate_type. Together they identify a row, so a caregiver has at most one charge of each type.
- **user_id** points to **caregiver_profiles.user_id**. Deleting that row also deletes this one.
- amount must be more than 0.

* the column is one part of a combination that has to be unique together.

4. caregiver_documents

The documents a caregiver uploaded (NIC, certificates) and the admin's decision on each. The files themselves are kept on the server's disk in a private folder; this table only holds the file name and the status. At most one current file per document type.

Column	Type	Empty allowed?	Key	Default	What it holds	Example value
id	bigint	No	PK	auto number	Unique number of the document row. Auto numbered.	7
user_id	bigint	No	FK, UQ*	-	The caregiver the document belongs to.	12
doc_type	varchar(30)	No	UQ*	-	Which document this is. Allowed words are on the last page.	NIC
file_name	varchar(80)	No	-	-	Generated name of the file in the private folder uploads/documents. Never shown to families.	8c41b0de-...-52.pdf
status	varchar(10)	No	-	-	The admin's decision. A new upload always starts as PENDING.	PENDING
revision	integer	No	-	1	Goes up by one every time a new file is uploaded in place of the old one. The admin's approve or reject must name the revision they looked at, so a file replaced meanwhile can never be approved unseen.	1
reject_reason	varchar(300)	Yes	-	-	Why the admin rejected it. The caregiver reads this. Empty unless rejected.	The photo is blurry
reviewed_by	bigint	Yes	FK	-	The admin (users.id) who approved or rejected it. Becomes empty if that admin account is deleted.	3
reviewed_at	timestamptz	Yes	-	-	When the admin decided. Empty while pending.	2026-10-28 11:40:00+05:30
uploaded_at	timestamptz	No	-	-	When the current file was uploaded.	2026-10-28 10:30:00+05:30

Links, rules and indexes

- **Primary key**: id.
- **reviewed_by** points to **users.id**. If that row is deleted, this value becomes empty.
- **user_id** points to **caregiver_profiles.user_id**. Deleting that row also deletes this one.
- A caregiver has at most one current file per document type (user_id + doc_type must be unique).
- **Index** `idx_caregiver_documents_status` on (status, uploaded_at): the admin's queue of documents waiting for review.

* the column is one part of a combination that has to be unique together.

5. document_requirements

Which document the admin made mandatory for which job. A job's document is optional unless a row here says required. Changing a row shows or hides the caregivers of that job straight away.

Column	Type	Empty allowed?	Key	Default	What it holds	Example value
job_type	varchar(30)	No	PK*	-	The job. Together with doc_type it forms the primary key.	CARETAKER
doc_type	varchar(30)	No	PK*	-	The document. It must be one that job can take.	NIC
required	boolean	No	-	false	True = mandatory: caregivers of this job without it approved are hidden. False, or no row at all = optional.	false
updated_at	timestamptz	No	-	-	When the admin last changed this setting.	2026-10-29 18:58:21+05:30
updated_by	bigint	Yes	FK	-	The admin (users.id) who changed it. Becomes empty if that account is deleted.	3

Links, rules and indexes

- **Primary key**: job_type, doc_type.
- **updated_by** points to **users.id**. If that row is deleted, this value becomes empty.

* the column is one part of a combination that has to be unique together.

6. reviews

A care seeker's star rating and comment for a caregiver. A seeker can review a caregiver only once: reviewing again replaces the first review.

Column	Type	Empty allowed?	Key	Default	What it holds	Example value
id	bigint	No	PK	auto number	Unique number of the review. Auto numbered.	5
caregiver_id	bigint	No	FK, UQ*	-	The caregiver being reviewed (caregiver_profiles.user_id).	12
reviewer_id	bigint	No	FK, UQ*	-	The care seeker who wrote it (users.id).	9
rating	integer	No	-	-	Stars, 1 to 5.	5
comment	varchar(1000)	Yes	-	-	Optional text, up to 1000 characters.	Wonderful and kind
created_at	timestamptz	No	-	-	When the first review was written.	2026-10-28 12:00:00+05:30
updated_at	timestamptz	No	-	-	When it was last changed. Equals created_at until the seeker edits it.	2026-10-28 12:00:00+05:30

Links, rules and indexes

- **Primary key**: id.
- **caregiver_id** points to **caregiver_profiles.user_id**. Deleting that row also deletes this one.
- **reviewer_id** points to **users.id**. Deleting that row also deletes this one.
- A seeker has only one review per caregiver (caregiver_id + reviewer_id must be unique).
- rating must be from 1 to 5.
- **Index** idx_reviews_caregiver on (caregiver_id, updated_at DESC): the reviews of one caregiver, newest first.

* the column is one part of a combination that has to be unique together.

7. bookings

A care seeker's request for a caregiver. The admin gets an alert, calls the seeker and moves the request along (requested, confirmed, completed or cancelled). When a way of paying is switched on, the admin also sets our fee, and the caregiver's own charge if it is collected too, when marking the visit completed; the seeker pays the total through PayHere or by bank transfer (see payments). The app never holds card details.

Column	Type	Empty allowed?	Key	Default	What it holds	Example value
id	bigint	No	PK	auto number	Request number shown to the admin and in the alert. Auto numbered.	4
seeker_id	bigint	No	FK	-	The care seeker who asked (users.id).	9
caregiver_id	bigint	No	FK	-	The caregiver they chose (caregiver_profiles.user_id).	12
status	varchar(20)	No	-	-	Where the request stands. Allowed words and the allowed moves are on the last page.	REQUESTED
seeker_phone	varchar(30)	No	-	-	The phone number the seeker gave for this request, so the admin can call them.	+94 77 123 4567
rate_type	varchar(20)	Yes	-	-	Optional. The way of charging the seeker picked.	DAILY
quoted_amount	numeric(10,2)	Yes	-	-	Optional. The caregiver's charge at the moment of the request. It is copied, so later price changes do not alter it.	9000.00
note	varchar(500)	Yes	-	-	Optional message from the seeker, up to 500 characters.	Mornings only
created_at	timestamptz	No	-	-	When the request was made.	2026-10-28 16:10:34+05:30
updated_at	timestamptz	No	-	-	When the status last changed.	2026-10-28 16:20:00+05:30
address	varchar(300)	Yes	-	-	Where the care is needed. Required when the seeker makes a request; the admin sees it. Empty for older requests.	5 Lake Road, Colombo
fee_amount	numeric(10,2)	Yes	-	-	Optional. Our fee, typed by the admin when marking the visit completed while a way of paying is switched on. The seeker pays it together with caregiver_amount. Empty for older requests and when nothing was switched on.	450.00
fee_requested_at	timestamptz	Yes	-	-	When the amounts were set or last changed (the seeker is emailed then).	2026-10-29 20:00:31+05:30
caregiver_amount	numeric(10,2)	Yes	-	-	Optional. The caregiver's own charge, collected from the seeker together with fee_amount (the admin then pays the caregiver). Empty when only the fee is collected.	9000.00

Links, rules and indexes

- **Primary key**: id.
- **caregiver_id** points to **caregiver_profiles.user_id**. Deleting that row also deletes this one.
- **seeker_id** points to **users.id**. Deleting that row also deletes this one.
- **Index** `idx_bookings_seeker` on (seeker_id, created_at DESC): a seeker's own list of requests, newest first.
- **Index** `idx_bookings_status` on (status, created_at DESC): the admin's list of requests by status, newest first.

8. payments

Every try to pay a request's amount. The seeker pressing Pay now makes a row; PayHere's signed answer then marks it paid, failed or cancelled. A failed try is followed by a new row, and the fee counts as paid once any row is PAID. The admin can also record a payment made outside PayHere (a bank transfer or cash). Card numbers are never stored.

Column	Type	Empty allowed?	Key	Default	What it holds	Example value
id	bigint	No	PK	auto number	Unique number of the payment try. Auto numbered.	2
booking_id	bigint	No	FK	-	The request this fee is for (bookings.id). Removed together with the request.	4
order_id	varchar(40)	No	UQ	-	What PayHere calls this try. A new one is made every time the seeker presses Pay now, so a failed try never mixes with the next. Must be unique.	HC4-CFA7D27691
amount	numeric(10,2)	No	-	-	The total asked for in this try (our fee plus the caregiver's charge), copied from the request at that moment. PayHere's answer must match it.	9450.00
currency	varchar(3)	No	-	-	The currency. Always LKR for now.	LKR
status	varchar(12)	No	-	-	How this try stands. Allowed words are on the last page.	PAID
source	varchar(10)	No	-	-	PAYHERE for an online payment, MANUAL when the admin recorded a payment made outside PayHere (bank deposit, cash).	PAYHERE
method	varchar(20)	Yes	-	-	How it was paid, as PayHere reports it (VISA, MASTER and so on), SIMULATED in the test simulator, OFFLINE for a manual one. Empty until known.	VISA
payhere_payment_id	varchar(40)	Yes	-	-	PayHere's own number for the payment, for finding it in the PayHere dashboard. Empty until PayHere answers.	320032145678
status_code	integer	Yes	-	-	PayHere's last status number: 2 paid, 0 pending, -1 cancelled, -2 failed, -3 charged back. Empty until PayHere answers.	2
created_at	timestamptz	No	-	-	When the seeker pressed Pay now (or the admin recorded the payment).	2026-10-29 20:01:20+05:30
updated_at	timestamptz	No	-	-	When PayHere's answer last changed this row.	2026-10-29 20:01:26+05:30
paid_at	timestamptz	Yes	-	-	When it was paid. Empty if it never was.	2026-10-29 20:01:26+05:30

Links, rules and indexes

- **Primary key**: id.
- **booking_id** points to **bookings.id**. Deleting that row also deletes this one.
- Every payment try has its own order id: no two can share one.
- **Index** `idx_payments_booking` on (booking_id): the payment tries of one request.

9. payment_settings

The admin's two switches for how care seekers may pay: card payments through PayHere and bank transfer. There is only one row. With both switched off, completing a visit asks for no fee.

Column	Type	Empty allowed?	Key	Default	What it holds	Example value
id	bigint	No	PK	-	Always 1: there is a single row of settings.	1
payhere_enabled	boolean	No	-	-	The admin's switch for card payments through PayHere. It only takes effect when the server is set up for PayHere.	true
bank_enabled	boolean	No	-	-	The admin's switch for bank transfers. It only takes effect when the bank details are set in the server's settings.	false
updated_at	timestamptz	No	-	-	When a switch was last changed.	2026-10-30 08:40:00+05:30
updated_by	bigint	Yes	FK	-	The admin who last changed a switch (users.id). Empty if it was never changed or that admin was deleted.	7

Links, rules and indexes

- **Primary key**: id.
- **updated_by** points to **users.id**. If that row is deleted, this value becomes empty.
- id must be 1: the table holds a single row of settings.

10. payment_receipts

The receipts care seekers send after paying by bank transfer: a screenshot or PDF per row, kept on the server's private disk (not a public address), at most 5 per request. Each one is also emailed to the admin with the file attached. A receipt never marks anything paid: the admin does that after checking the bank account.

Column	Type	Empty allowed?	Key	Default	What it holds	Example value
id	bigint	No	PK	auto number	Unique number of the receipt. Auto numbered.	3
booking_id	bigint	No	FK	-	The request the receipt is for (bookings.id). Removed together with the request.	12
file_name	varchar(80)	No	-	-	Generated name of the file in the server's private receipts folder. Only an admin can open it, through the Admin page.	9f1c2d7e-...-e2.png
note	varchar(300)	Yes	-	-	Optional. What the seeker wrote with the receipt, up to 300 characters.	Paid from HNB
uploaded_at	timestamptz	No	-	-	When the receipt was sent.	2026-10-30 09:10:00+05:30

Links, rules and indexes

- **Primary key**: id.
- **booking_id** points to **bookings.id**. Deleting that row also deletes this one.
- **Index** idx_payment_receipts_booking on (booking_id): the receipts of one request.

11. notifications

The queue of emails and WhatsApp alerts. When a booking is saved an alert for the admin is saved with it; completing a visit with a fee queues a 'please pay' email to the seeker, and a payment queues receipts. A background job sends them and retries on failure. The 'params' column holds names, phones and emails, so treat this table as personal data.

Column	Type	Empty allowed?	Key	Default	What it holds	Example value
id	bigint	No	PK	auto number	Unique number of the alert. Auto numbered.	6
booking_id	bigint	Yes	FK	-	The request this alert is about. Empty for alerts that are not about a request.	4
channel	varchar(20)	No	-	-	How the alert goes out. Allowed words are on the last page. Payment emails always go by EMAIL.	EMAIL
recipient	varchar(500)	No	-	-	Where it goes: the admin's email address (several separated by commas), a WhatsApp number, or the care seeker's email for payment emails. 'not-set' if none was configured.	admin@example.com
template	varchar(80)	No	-	-	Name of the message: new_booking_details is the admin's alert; fee_due, payment_receipt, payment_received and receipt_uploaded are the payment emails.	new_booking_details
language	varchar(10)	No	-	-	Language code of the template.	en
params	varchar(4000)	No	-	-	JSON list of the values filled into the message, in order. For a new request: request number, seeker name, phone, email, care address, caregiver, job, charge, note. The payment emails have shorter lists.	["#4","Sam Seeker","+94 77 123 4567", ...]
status	varchar(12)	No	-	-	Whether the alert went out. Allowed words are on the last page.	SENT
attempts	integer	No	-	0	How many times sending was tried. It stops after 5.	1
last_error	varchar(500)	Yes	-	-	The reason the last attempt failed. Empty when it worked.	The mail server refused the username or password.
next_attempt_at	timestamptz	No	-	-	The earliest time to try sending again. The background job picks up alerts that are due.	2026-10-28 16:10:34+05:30
created_at	timestamptz	No	-	-	When the alert was queued.	2026-10-28 16:10:34+05:30
sent_at	timestamptz	Yes	-	-	When it was sent. Empty until sent.	2026-10-28 16:10:40+05:30

Links, rules and indexes

- **Primary key**: id.
- **booking_id** points to **bookings.id**. Deleting that row also deletes this one.
- **Index** idx_notifications_due on (status, next_attempt_at): finding the alerts that are due to be sent.
- **Index** idx_notifications_booking on (booking_id): the alerts of one request.

Allowed values of the fixed-word columns

These columns are saved as plain text. The application writes only the words below. The database itself rejects other words only for users.role.

Column	Allowed values	Meaning
`users.role`	CAREGIVER, CARE_SEEKER	Kind of account. Admin is not a role: it is the separate flag users.is_admin.
`caregiver_profiles.job_type`	CARETAKER, NURSING, PHYSIOTHERAPIST	The work a caregiver does. Care seekers browse by this.
`caregiver_rates.rate_type and bookings.rate_type`	HOURLY, SESSION, DAILY, FULL_DAY_NIGHT	Per hour, per session, per day, full day and night.
`bookings.status`	REQUESTED, CONFIRMED, COMPLETED, CANCELLED	A request starts as REQUESTED. The admin can move it REQUESTED to CONFIRMED or CANCELLED, and CONFIRMED to COMPLETED or CANCELLED. A seeker can cancel only while it is still REQUESTED. COMPLETED and CANCELLED are final.
`notifications.channel`	EMAIL, WHATSAPP	How the alert goes out. One channel at a time, chosen in the settings.
`notifications.status`	PENDING, SENT, DRY_RUN, FAILED	PENDING: waiting to be sent or retried. SENT: delivered to the mail server or WhatsApp. DRY_RUN: test mode, only written to the log. FAILED: gave up after 5 tries, or a permanent error.
`notifications.language`	en	Language code of the message template.
`notifications.template`	new_booking_details, fee_due, payment_receipt, payment_received, receipt_uploaded	Which message it is. The first is the admin's alert for a new request; the other four are the payment emails.
`payments.status`	PENDING, PAID, FAILED, CANCELED, CHARGEDBACK	PENDING: started, PayHere has not answered yet. PAID: the money was received. FAILED: the card was refused. CANCELED: the seeker cancelled, or the admin changed the fee so this try no longer counts. CHARGEDBACK: the bank reversed a payment. Only PayHere's signed message, or the admin recording a payment, can make a row PAID.
`payments.source`	PAYHERE, MANUAL	PAYHERE: paid online. MANUAL: the admin recorded a payment made outside PayHere.
`payments.currency`	LKR	Sri Lankan rupees, the only currency used for now.
`caregiver_documents.doc_type`	NIC, DEGREE_CERTIFICATE, NURSING_CERTIFICATE, GS_LETTER	See the table below for which job can take which.
`caregiver_documents.status`	PENDING, APPROVED, REJECTED	A new upload is PENDING. The admin approves it, or rejects it with a reason.
`users.is_admin, caregiver_profiles.available, caregiver_profiles.listed, document_requirements.required, payment_settings.payhere_enabled, payment_settings.bank_enabled`	true, false	Yes/no switches.

Which documents each job can take

All of them are optional until the admin makes one mandatory for a job (table document_requirements).

caregiver_profiles.job_type	Documents the caregiver can upload
Caretaker	NIC, Grama Niladhari (GS) letter
Nursing	Nursing qualification certificate, NIC
Physiotherapist	Degree certificate, NIC